Description of files in this folder

1. generating_counts_table.md
    a. File describing how the counts table (peaks x samples) was generated from BAM files.
        i. The BAM files were produced from the raw FASTQ files available on SRA, as described in Supplementary Note 2.
    b. There are three main steps here, as described here (PMID 33606259)
        i. Peak calling and IDR to determine confident peaks per sample
        ii. Merging peaks across samples
        iii. FeatureCounts to generate the counts table
    c. Please note, the markdown mentions/describes the following files that are also contained in this folder:
        i. peak_merge_atac_env.yml
        ii. for_merging_peaks subfolder
        iii. merged_peaks subfolder
        iv. iterative_overlap_metadata.txt
        v. All_Samples.counts
2. deseq2.Rmd
    a. Input:
        i. The All_Samples.counts file generated in the previous step and contained in this folder
    b. Outputs:
        i. Volcano plot used for a figure panel
        ii. DESeq2 results (deseq_results.tsv) contained in this folder
            1. Also deseq_results_with_brain3.tsv (results that include the sample with low TSS enrichment score that was excluded from the analyses in the paper, see Supplementary Note 2)
        iii. Files needed for downstream analyses, see below
            1. sig_peak_file_for_homer.txt
            2. up_peaks_input_for_cov_heatmaps.bed
            3. down_peaks_input_for_cov_heatmaps.bed
3. homer_command.sh
    a. Command used to run HOMER to annotate the significantly differentially accessible peaks
        i. Requires the sig_peak_file_for_homer.txt (generated above)
4. differential_peak_annotation_analysis.Rmd
    a. Inputs
        i. Outputs from HOMER, contained within this folder
            1. annotation_stats.txt

            2. annotated_sig_peak_file.txt
        ii. deseq_results.tsv (generated above)
        iii. DESeq2 results from the mouse brain RNA-seq experiment for GDCA
            1. Contained within the “Mouse brain RNAseq” folder of this repo
    b. Outputs
        i. A pie chart showing the percentage of differentially accessible peaks annotated to each genomic category, such as introns, exons, and promoters, for use in a figure.
        ii. A bar chart showing whether each genomic category is enriched or depleted among differentially accessible peaks relative to the genomic background, for use in a figure.
        iii. A text file - for use in a supplementary table - containing HOMER and rGREAT annotations for each differentially accessible peak. The file also indicates whether genes linked to each peak by rGREAT showed differential expression in the RNA-seq analysis.
5. generating_cov_matrices.md
    a. Markdown file describing the process that was used to generate the coverage matrices that were input into cov_heatmap.Rmd (below)
6. cov_heatmap.Rmd
    a. Inputs:
        i. The four coverage matrices generated in the previous step
    b. Outputs:
        i. Coverage heatmaps in saline and GDCA samples for peaks identified as differentially accessible between saline- and GDCA-treated brains.
7. chromvar.Rmd
    a. Inputs
        i. All_Samples.counts generated above
    b. Outputs – all contained within chromvar_results subfolder
        i. A text file showing variability scores across tested samples, per TF motif
            1. Incorporated into a supplementary table
        ii. Heatmap showing correlations between samples based on TF motif accessibility
            1. Used for a figure
        iii. Heatmap of chromVAR deviation score (proxy for accessibility) for the 25 TF motifs with the greatest variability in accessibility across samples

        1. Used for a figure

iv. Plot showing the variability estimates for the 10 TF motifs with the greatest variability in accessibility across samples

    1. Used for a figure